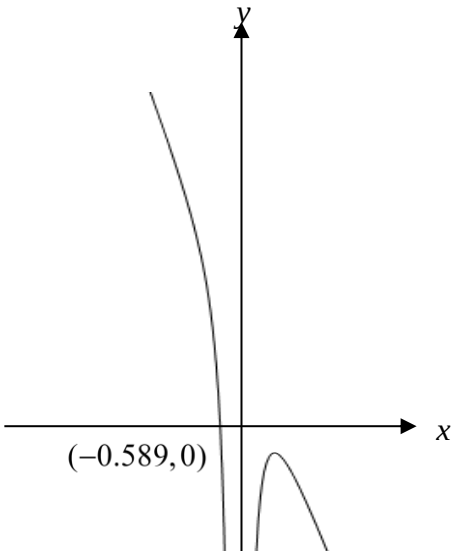
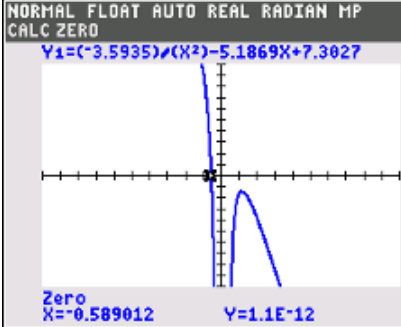
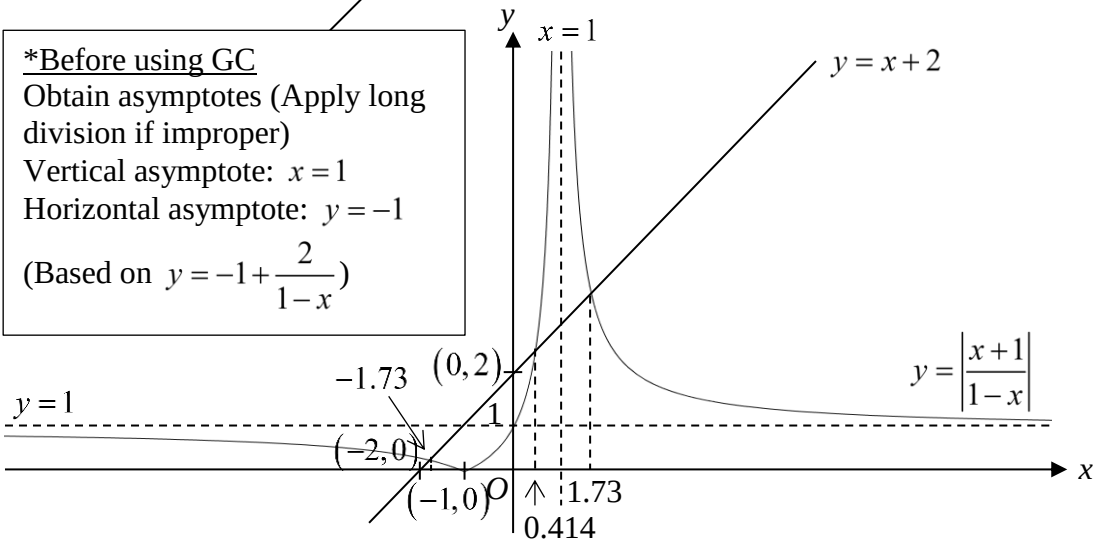
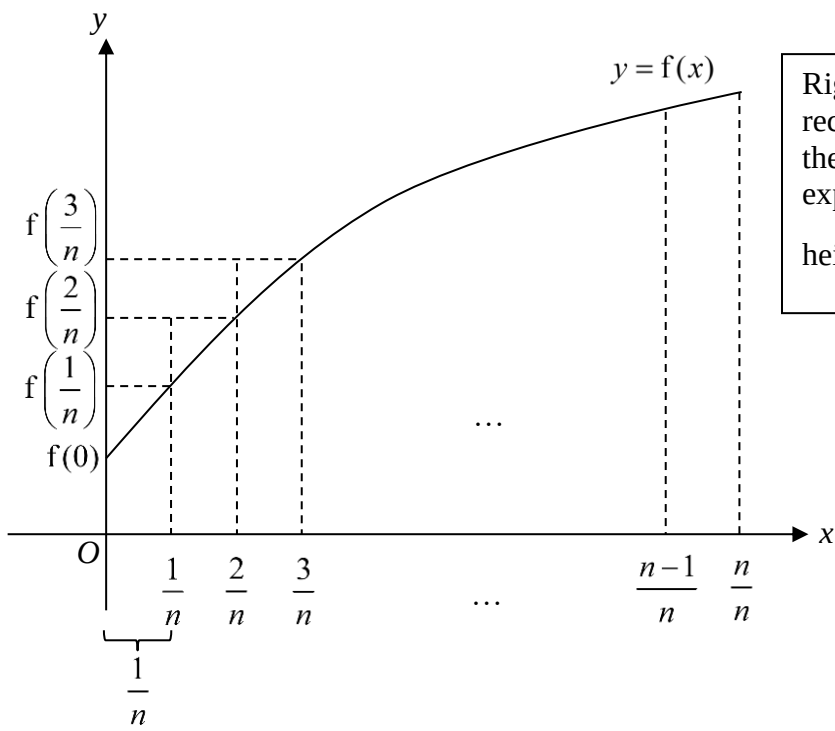


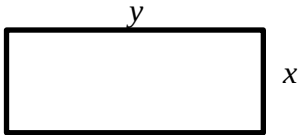
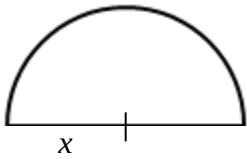
2015 A Level Paper 1 Suggested Solutions

1	Suggested Solutions												
(i)	$\frac{dy}{dx} = -2ax^{-3} + b \quad \text{--- (*)}$ Substitute $x = 1$ into (*), $\frac{dy}{dx} = -2a(1)^{-3} + b = 2$$-2a + b = 2 \quad \text{--- (1)}$ Substitute $(1.6, -2.4)$ into $y = \frac{a}{x^2} + bx + c$, $\frac{1}{1.6^2}a + 1.6b + c = -2.4 \quad \text{--- (2)}$ Substitute $(-0.7, 3.6)$ into $y = \frac{a}{x^2} + bx + c$, $\frac{1}{(-0.7)^2}a - 0.7b + c = 3.6 \quad \text{--- (3)}$ Solving (1), (2) and (3) using GC, $\left. \begin{aligned} a &= -3.59345 \approx -3.593 \\ b &= -5.18691 \approx -5.187 \\ c &= 7.30274 \approx 7.303 \end{aligned} \right\} \text{correct to 3 d.p.}$ NORMAL FLOAT AUTO REAL DEGREE MP PLYSMLT2 APP SYSTEM OF EQUATIONS <table><tr><td>0.39...x+</td><td>1.6y+</td><td>1z=</td><td>-2.4</td></tr><tr><td>2.04...x+</td><td>-0.7y+</td><td>1z=</td><td>3.6</td></tr><tr><td>-2x+</td><td>1y+</td><td>0z=</td><td>2</td></tr></table> 2 MAIN MODE CLEAR LOAD SOLVE NORMAL FLOAT AUTO REAL DEGREE MP PLYSMLT2 APP SOLUTION x=-3.593453398 y=-5.186906795 z=7.302743606 MAIN MODE SYSTM STORE F < > D	0.39...x+	1.6y+	1z=	-2.4	2.04...x+	-0.7y+	1z=	3.6	-2x+	1y+	0z=	2
0.39...x+	1.6y+	1z=	-2.4										
2.04...x+	-0.7y+	1z=	3.6										
-2x+	1y+	0z=	2										
(ii)	 NORMAL FLOAT AUTO REAL RADIAN MP CALC ZERO $Y1=(-3.5935)/(X^2)-5.1869X+7.3027$  Zero X=-0.589012 Y=1.1E-12 Use 5 s.f. values of a, b and c. (intermediate step use 5 s.f.) Using GC, $x = -0.589$ (3 d.p.)												
(iii)	$y = -5.187x + 7.303$ $y = -5.19x + 7.30$ (correct to 3 s.f.)												

2	Suggested Solutions	
(i)	$y = \left \frac{x+1}{1-x} \right = \left \frac{-(1-x)+2}{1-x} \right = \left -1 + \frac{2}{1-x} \right $ <u>*Before using GC</u> Obtain asymptotes (Apply long division if improper) Vertical asymptote: $x = 1$ Horizontal asymptote: $y = -1$ (Based on $y = -1 + \frac{2}{1-x}$)	Sketch $y = \frac{x+1}{1-x}$ (finding all important features), then sketch graph of $y = f(x)$. 
(ii)	From graph, Using GC, $-1.73 < x < 0.414$ or $x > 1.73$ $\left \frac{x+1}{1-x} \right < x+2 \Rightarrow$ Find values of x where $y = \left \frac{x+1}{1-x} \right$ is <u>below</u> $y = x+2$.	

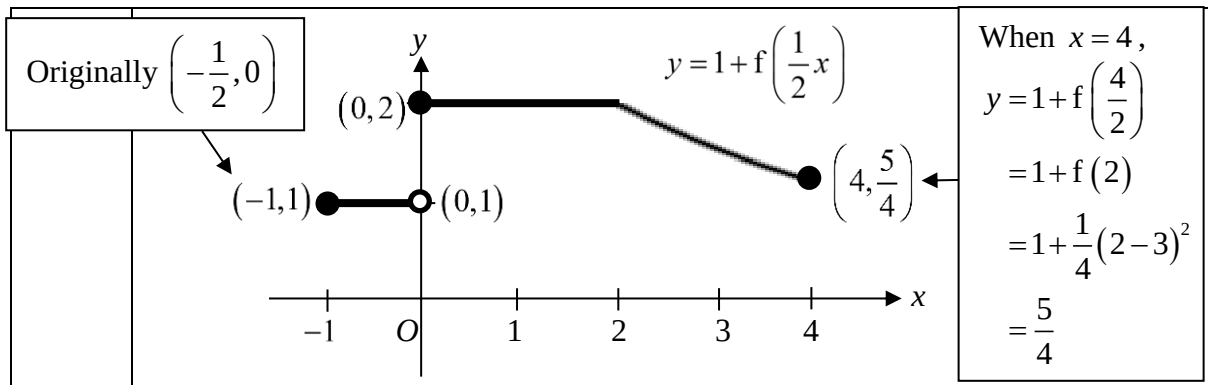
3	Suggested Solutions
(i)	 Right side of each rectangle must be the height as 1st expression for height is $f\left(\frac{1}{n}\right)$. Using integration, area bounded by $y = f(x)$, $x = 0$, $x = 1$ and the x-axis is given by $\int_0^1 f(x) dx$.

	Also, the same area can be estimated by first, dividing x values from 0 to 1 into n equal parts, forming n rectangles of equal width of $\frac{1}{n}$. Hence, the same area is estimated by finding the total areas of the rectangles $= \frac{1}{n} \times f\left(\frac{1}{n}\right) + \frac{1}{n} \times f\left(\frac{2}{n}\right) + \frac{1}{n} \times f\left(\frac{3}{n}\right) + \dots + \frac{1}{n} \times f\left(\frac{n-1}{n}\right) + \frac{1}{n} \times f\left(\frac{n}{n}\right)$ $= \frac{1}{n} \left\{ f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + f\left(\frac{3}{n}\right) + \dots + f\left(\frac{n}{n}\right) \right\}$ As the number of rectangles, n, increases to infinity, $\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + f\left(\frac{3}{n}\right) + \dots + f\left(\frac{n}{n}\right) \right\} = \int_0^1 f(x) \, dx \text{ (shown)}$
(ii)	$\lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{\sqrt[3]{1} + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{\sqrt[3]{n}} \right)$ $= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt[3]{\frac{1}{n}} + \sqrt[3]{\frac{2}{n}} + \dots + \sqrt[3]{\frac{n}{n}} \right) \leftarrow \text{Observe } f(x) = \sqrt[3]{x}$ $= \int_0^1 \sqrt[3]{x} \, dx$ $= \left. \frac{x^{\frac{4}{3}}}{\frac{4}{3}} \right _0^1$ $= \frac{3}{4}$

4	Suggested Solutions
	  $2(x + y) + 2x + \frac{1}{2} \times 2\pi x = d$ $2x + 2y + 2x + \pi x = d$ $y = \frac{d - \pi x - 4x}{2} \quad \text{--- (1)}$ Let A be the total area of rectangle and semicircle. $A = xy + \frac{1}{2} \pi x^2 \quad \text{--- (2)} \leftarrow$ Substitute (2) into (1): A is in terms of x and y (2 variables) $\Rightarrow$ Need to find an equation involving x and y so that we can eliminate 1 variable. Note: d is a constant (fixed length)

$A = x \left(\frac{d - \pi x - 4x}{2} \right) + \frac{1}{2} \pi x^2$ $A = \frac{d}{2} x - \frac{1}{2} \pi x^2 - 2x^2 + \frac{1}{2} \pi x^2$ $A = \frac{d}{2} x - 2x^2$ $\therefore \frac{dA}{dx} = \frac{d}{2} - 4x$ For maximum A, $\frac{dA}{dx} = 0 \Rightarrow \frac{d}{2} - 4x = 0 \Rightarrow x = \frac{d}{8}$ $\frac{d^2 A}{dx^2} = -4 < 0 \quad \therefore A \text{ is maximum when } x = \frac{d}{8}.$ $\therefore A = \frac{d}{2} \left(\frac{d}{8} \right) - 2 \left(\frac{d}{8} \right)^2 = \frac{d^2}{32} \text{ (shown)}$ $\therefore k = \frac{1}{32}$	Need to calculate value of $\frac{d^2 A}{dx^2}$ when $x = \frac{d}{8}$ if using 2nd derivative test. (Can use GC if no unknown)
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5	Suggested Solutions
(i)	$y = x^2 \xrightarrow{(1) \text{ replace } y \text{ by } \frac{y}{4}} y = \frac{x^2}{4} \xrightarrow{(2) \text{ replace } x \text{ by } x-3} y = \frac{1}{4}(x-3)^2$ (1) Scaling parallel to y-axis by a factor of $\frac{1}{4}$. (2) Translate 3 units in the positive x-direction. *Any order is accepted. <u>Alternative</u> (1) Scaling parallel to x-axis by a factor of 2. (2) Translate 3 units in the positive x-direction. Check transformation using this diagram first!
(ii)	Include coordinates of endpoints.
(iii)	$y = f(x) \xrightarrow{\text{replace } x \text{ by } \frac{x}{2}} y = f\left(\frac{x}{2}\right) \xrightarrow{\text{replace } y \text{ by } y-1} y = f\left(\frac{x}{2}\right) + 1$ x-coordinates $\times 2$ y-coordinates $+1$



6 Suggested Solutions	
(i)	$\ln(1+2x) = 2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} + \dots$ $= 2x - 2x^2 + \frac{8}{3}x^3 + \dots$ (MF26) Replace x by $2x$ for expansion of $\ln(1+x)$
(ii)	$ax(1+bx)^c = ax\left(1+bcx + \frac{c(c-1)}{2}(bx)^2 + \dots\right)$ $= ax + abcx^2 + \frac{c(c-1)}{2}ab^2x^3 + \dots$ Use expansion of $(1+x)^n$ in MF26 for $(1+bx)^c$ By comparing part (i)/(ii), $a = 2$ $abc = -2 \Rightarrow b = -\frac{1}{c} \quad (\because a = 2)$ $\frac{c(c-1)}{2}ab^2 = \frac{8}{3}$ As $b = -\frac{1}{c}$, $\frac{c(c-1)}{2} \times 2 \left(-\frac{1}{c}\right)^2 = \frac{8}{3}$ $c(c-1)\left(\frac{1}{c^2}\right) = \frac{8}{3}$ $(c-1)\left(\frac{1}{c}\right) = \frac{8}{3}$ $3c - 3 = 8c$ $c = -\frac{3}{5}$ $\therefore b = \frac{5}{3}$ Coefficient of $x^4 = a \left[\frac{c(c-1)(c-2)}{3!} \times b^3 \right] = 2 \left[\frac{-\frac{3}{5}\left(-\frac{8}{5}\right)\left(-\frac{13}{5}\right)}{3!} \left(\frac{5}{3}\right)^3 \right]$ $= -\frac{104}{27}$

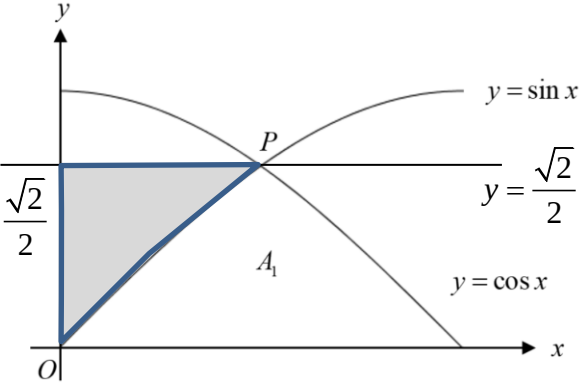
7	Suggested Solutions
(i)	From the diagram, $\overrightarrow{OC} = \frac{3}{5}\overrightarrow{OA} = \frac{3}{5}\mathbf{a}$, $\overrightarrow{OD} = \frac{5}{11}\overrightarrow{OB} = \frac{5}{11}\mathbf{b}$
(ii)	$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB}$ $= \frac{3}{5}\mathbf{a} - \mathbf{b}$ Equation of line BC: $\mathbf{r} = \overrightarrow{OB} + \lambda \overrightarrow{BC}$ $\mathbf{r} = \mathbf{b} + \lambda \left(\frac{3}{5}\mathbf{a} - \mathbf{b} \right)$ $\mathbf{r} = \frac{3}{5}\lambda \mathbf{a} + (1 - \lambda)\mathbf{b}, \lambda \in \mathbb{R} \text{ (shown)}$ $\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA}$ $= \frac{5}{11}\mathbf{b} - \mathbf{a}$ Equation of line AD: $\mathbf{r} = \overrightarrow{OA} + \mu \overrightarrow{AD}$ $\mathbf{r} = \mathbf{a} + \mu \left(\frac{5}{11}\mathbf{b} - \mathbf{a} \right)$ $\mathbf{r} = \frac{5}{11}\mu \mathbf{b} + (1 - \mu)\mathbf{a}, \mu \in \mathbb{R}$ Position vector of fixed point on line BC is $\overrightarrow{OB}$ or $\overrightarrow{OC}$ Direction vector of line BC is $\overrightarrow{BC}$ or $\overrightarrow{CB}$. Position vector of fixed point on line AD is $\overrightarrow{OA}$ or $\overrightarrow{OD}$ Direction vector of line AD is $\overrightarrow{AD}$ or $\overrightarrow{DA}$.
(iii)	Position vector of E, intersection of AD and BC $= \frac{3}{5}\lambda \mathbf{a} + (1 - \lambda)\mathbf{b} = \frac{5}{11}\mu \mathbf{b} + (1 - \mu)\mathbf{a}$ Since $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel vectors, $\frac{3}{5}\lambda = 1 - \mu \quad \text{or} \quad 1 - \lambda = \frac{5}{11}\mu \quad \text{--- (2)}$ $\mu = 1 - \frac{3}{5}\lambda \quad \text{--- (1)}$ Solving (1) and (2), $1 - \lambda = \frac{5}{11} \left(1 - \frac{3}{5}\lambda \right)$ $1 - \lambda = \frac{5}{11} - \frac{3}{11}\lambda$ $\lambda = \frac{3}{4}$ Recall: For $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non parallel vectors, $\lambda \mathbf{a} = \mu \mathbf{b} \Rightarrow \lambda = \mu = 0$

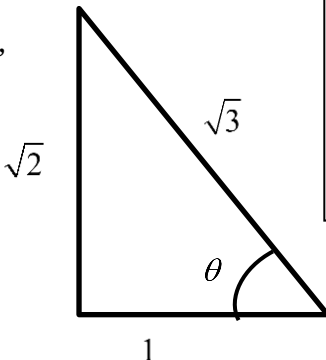
	$\therefore \overrightarrow{OE} = \frac{3}{5} \left(\frac{3}{4} \right) \mathbf{a} + \left(1 - \frac{3}{4} \right) \mathbf{b}$ $= \frac{9}{20} \mathbf{a} + \frac{1}{4} \mathbf{b}$ $\overrightarrow{AE} = \overrightarrow{OE} - \overrightarrow{OA}$ $= \frac{9}{20} \mathbf{a} + \frac{1}{4} \mathbf{b} - \mathbf{a}$ $= \frac{1}{4} \mathbf{b} - \frac{11}{20} \mathbf{a}$ $= \frac{11}{20} \left(\frac{5}{11} \mathbf{b} - \mathbf{a} \right)$ $= \frac{11}{20} \overrightarrow{AD}$ $\therefore AE : ED \Rightarrow 11 : 9$
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8	Suggested Solutions
(i)	$T, T+2, T+4, \dots$ (AP: 1st term, $a = T$, common difference, $d = 2$) $1\frac{1}{2} \text{ h} = 5400 \text{ secs}$, $1\frac{3}{4} \text{ h} = 6300 \text{ secs}$ *Understand the question by <u>LISTING</u> the terms. $5400 \leq S_{50} \leq 6300$ $5400 \leq \frac{50}{2} [2T + (50-1)(2)] \leq 6300$ $5400 \leq \frac{50}{2} [2T + 49(2)] \leq 6300$ $108 \leq T + 49 \leq 126$ $59 \leq T \leq 77$ $\therefore \{T \in \mathbb{R} : 59 \leq T \leq 77\}$ <u>Recall:</u> Sum of AP, $S_n = \frac{n}{2} [2a + (n-1)d]$
(ii)	$t, 1.02t, (1.02)^2 t, \dots$ (GP: 1st term, $a = t$, common ratio, $r = 1.02$) $5400 \leq S_{50} \leq 6300$ $5400 \leq \frac{t(1.02^{50} - 1)}{1.02 - 1} \leq 6300$ $108 \leq t(1.02^{50} - 1) \leq 126$ $63.845 \leq t \leq 74.486$ $\therefore \{t \in \mathbb{R} : 63.9 \leq T \leq 74.4\}$ (round off to 3 s.f.) <u>Recall:</u> Sum of GP, $S_n = \frac{a(1-r^n)}{1-r}$ or $\frac{a(r^n-1)}{r-1}$
(iii)	Athlete A: 50th lap $= a + (n-1)d = 59 + 49(2) = 157$ (50th term of AP) Athlete B: 50th lap $= ar^{n-1} = 63.845(1.02^{49}) = 168.47$ (50th term of GP) Difference between Athlete A and Athlete B $= 168.47 - 157$ $= 11.47$ $\approx 11 \text{ secs}$ (correct to nearest second)

9	Suggested Solutions
(a)	$\frac{w^2}{w^*} = \frac{(a+ib)^2}{a-ib}$ $= \frac{(a+ib)^3}{(a-ib)(a+ib)}$ $= \frac{a^3 + 3a^2bi - 3ab^2 - b^3i}{a^2 + b^2}$ $= \frac{a^3 - 3ab^2}{a^2 + b^2} + \frac{3a^2b - b^3}{a^2 + b^2}i$ For $\frac{w^2}{w^*}$ is purely imaginary, $\Rightarrow \frac{a^3 - 3ab^2}{a^2 + b^2} = 0$ $a^3 - 3ab^2 = 0$ $a(a - \sqrt{3}b)(a + \sqrt{3}b) = 0$ $a = 0 \text{ (rejected as } a \neq 0) \text{ or } a = \sqrt{3}b \text{ or } a = -\sqrt{3}b$ $b = \frac{1}{\sqrt{3}}a \quad b = -\frac{1}{\sqrt{3}}a$ $\therefore w = a + i\left(\pm \frac{1}{\sqrt{3}}a\right) = \left(1 \pm \frac{1}{\sqrt{3}}i\right)a$ <u>Aim:</u> Given $\frac{w^2}{w^*}$ is purely imaginary $\Rightarrow$ Express $\frac{w^2}{w^*} = x + yi$ and solve $\text{Re}\left(\frac{w^2}{w^*}\right) = x = 0$. - “Realise” (Rationalise) denominator in order to express $\frac{w^2}{w^*} = x + yi$. - Simplify powers of i using $i^2 = -1$.
(b)(i)	[Out of syllabus (9758)]
(b)(ii)	[Out of syllabus (9758)]

10	Suggested Solutions	Split into 2 areas
(i)	$A_1 = \int_0^{\frac{\pi}{4}} \sin x \, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x \, dx$ $= [-\cos x]_0^{\frac{\pi}{4}} + [\sin x]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ $= -\cos \frac{\pi}{4} + \cos 0 + \sin \frac{\pi}{2} - \sin \frac{\pi}{4}$ $= 2 - \sqrt{2}$ $A_2 = \int_0^{\frac{\pi}{4}} (\cos x - \sin x) \, dx$ $= [\sin x + \cos x]_0^{\frac{\pi}{4}}$ $= \left(\sin \frac{\pi}{4} + \cos \frac{\pi}{4}\right) - (\sin 0 + \cos 0)$ $= \sqrt{2} - 1$ Top curve – bottom curve	

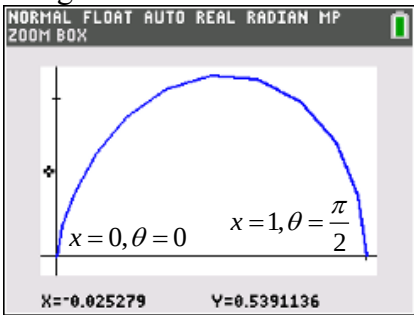
	$\therefore \frac{A_1}{A_2} = \frac{2-\sqrt{2}}{\sqrt{2}-1} = \frac{(2-\sqrt{2})(\sqrt{2}+1)}{(\sqrt{2}-1)(\sqrt{2}+1)} = \frac{2\sqrt{2}-2+2-\sqrt{2}}{(\sqrt{2})^2-1^2} = \frac{\sqrt{2}}{1} = \sqrt{2} \text{ (shown)}$	
(ii)	 $y = \sin x$ $\sin^{-1} y = x$ $\therefore$ Volume of the solid formed $= \pi \int_a^b x^2 dy = \pi \int_0^{\frac{\sqrt{2}}{2}} (\sin^{-1} y)^2 dy$ (shown)	Rotate about Y-AXIS $\Rightarrow \pi \int_{y_1}^{y_2} x^2 dy$ 1) Express x in terms of y. 2) Limits must be y values.
(iii)	$\pi \int_0^{\frac{\sqrt{2}}{2}} (\sin^{-1} y)^2 dy$ $= \pi \int_0^{\frac{\pi}{4}} [\sin^{-1}(\sin u)]^2 \cos u du$ $= \pi \int_0^{\frac{\pi}{4}} u^2 \cos u du \text{ (shown)}$	$y = \sin u$ $\frac{dy}{du} = \cos u$ when $y = 0, u = 0$. when $y = \frac{\sqrt{2}}{2}, u = \frac{\pi}{4}$. Integration by substitution. Change all y into u. 1) Change limits, $y = 0, y = \frac{\sqrt{2}}{2}$ 2) Change expression, $(\sin^{-1} y)^2$ 3) Change dy.
	Based on LIATE to choose U, $U = u^2$ (Algebraic), $V = \cos u$ (Trigo)	
	$\therefore \pi \int_0^{\frac{\pi}{4}} u^2 \cos u du$ $= \pi \left(u^2 \sin u \Big _0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} 2u \sin u du \right)$ $= \pi \left[\frac{\sqrt{2}\pi^2}{32} - 2 \left(-u \cos u \Big _0^{\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} \cos u du \right) \right]$ $= \pi \left[\frac{\sqrt{2}\pi^2}{32} - 2 \left(-\frac{\sqrt{2}\pi}{8} + [\sin u]_0^{\frac{\pi}{4}} \right) \right]$ $= \pi \left[\frac{\sqrt{2}\pi^2}{32} - 2 \left(-\frac{\sqrt{2}\pi}{8} + \frac{\sqrt{2}}{2} \right) \right]$ $= \pi \left(\frac{\sqrt{2}\pi^2}{32} + \frac{\sqrt{2}\pi}{4} - \sqrt{2} \right)$ $= \sqrt{2}\pi \left(\frac{\pi^2}{32} + \frac{\pi}{4} - 1 \right) \text{ units}^2$	Integration by parts, $\nearrow \text{KI} - \int \text{DI} dx \nwarrow$ <u>Keep U, Integrate V</u> <u>Differentiate U, Integrate V</u>

Qn 11	Suggested Solutions
(i)	$x = \sin^3 \theta, \quad y = 3\sin^2 \theta \cos \theta$ $\frac{dx}{d\theta} = 3\sin^2 \theta \cos \theta$ $\frac{dy}{d\theta} = 3[2\sin \theta \cos \theta \cos \theta + \sin^2 \theta (-\sin \theta)] = 3(2\sin \theta \cos^2 \theta - \sin^3 \theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3(2\sin \theta \cos^2 \theta - \sin^3 \theta)}{3\sin^2 \theta \cos \theta}$ $= \frac{2\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta}$ $= \frac{2\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$ $= 2\cot \theta - \tan \theta \quad (\text{shown})$
(ii)	At turning point, $\frac{dy}{dx} = 0$ $2\cot \theta - \tan \theta = 0$ $\frac{2}{\tan \theta} - \tan \theta = 0$ $2 - \tan^2 \theta = 0$ $\tan^2 \theta = 2$ $\tan \theta = \pm\sqrt{2}$ $\tan \theta = \sqrt{2} \quad \left(\because \tan \theta \text{ is positive as } 0 \leq \theta \leq \frac{\pi}{2} \right) \quad (\text{shown})$ $\therefore k = 2$ Since $\tan \theta = \sqrt{2}$ and $0 \leq \theta \leq \frac{\pi}{2}$, $\cos \theta = \frac{1}{\sqrt{3}}$ $\sin \theta = \sqrt{\frac{2}{3}}$  <u>Alternative</u> Use Trigo Identities $\sin^2 \theta + \cos^2 \theta = 1$ $\tan^2 \theta + 1 = \sec^2 \theta$ $\cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$ $\therefore x = \left(\sqrt{\frac{2}{3}} \right)^3 = \frac{2\sqrt{6}}{9}, \quad y = 3 \left(\sqrt{\frac{2}{3}} \right)^2 \left(\frac{1}{\sqrt{3}} \right) = \frac{2}{\sqrt{3}}$

To show maximum point			
x	$\left(\frac{2\sqrt{6}}{9}\right)^{-}$	$\frac{2\sqrt{6}}{9}$	$\left(\frac{2\sqrt{6}}{9}\right)^{+}$
$\frac{dy}{dx}$	+	0	-
Slope	$\nearrow$	$\longrightarrow$	$\searrow$

Alternative (Note: Finding 2nd derivative for parametric equations is out of syllabus for 9758, but you can use this if you know)

$$\begin{aligned}
 \frac{d^2y}{dx^2} &= (-2\operatorname{cosec}^2\theta - \sec^2\theta) \frac{d\theta}{dx} \\
 &= \left(-\frac{2}{\sin^2\theta} - \frac{1}{\cos^2\theta}\right) \frac{1}{3\sin^2\theta \cos\theta} \\
 &= \left(-\frac{2}{\left(\frac{\sqrt{2}}{\sqrt{3}}\right)^2} - \frac{1}{\left(\frac{1}{\sqrt{3}}\right)^2}\right) \frac{1}{3\left(\frac{\sqrt{2}}{\sqrt{3}}\right)^2 \frac{1}{\sqrt{3}}} \\
 &= (-3-3) \frac{\sqrt{3}}{2} \\
 &= -3\sqrt{3} < 0 \\
 \therefore \text{At } \left(\frac{2\sqrt{6}}{9}, \frac{2}{\sqrt{3}}\right), &\text{ it is a maximum point.}
 \end{aligned}$$

(iii)	When $x = 0, \theta = 0$ When $x = 1, \theta = \frac{\pi}{2}$ $\frac{dx}{d\theta} = 3\sin^2\theta \cos\theta$ Required area $= \int_0^1 y \, dx$ $= \int_0^{\frac{\pi}{2}} 3\sin^2\theta \cos\theta (3\sin^2\theta \cos\theta) \, d\theta$ $= \int_0^{\frac{\pi}{2}} 9\sin^4\theta \cos^2\theta \, d\theta \quad (\text{shown})$ Using GC, $\int_0^{\frac{\pi}{2}} 9\sin^4\theta \cos^2\theta \, d\theta = 0.884 \text{ units}^2$ (correct to 3 d.p.)	-Always use GC to sketch the graph to identify region. - Remember to set values of T using "WINDOWS" 
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(iv)	$x = \sin^3\theta, y = 3\sin^2\theta \cos\theta$ ----- (1) Since $y = ax$ ----- (2) Substitute (1) into (2)	Finding coordinates of intersections $\Rightarrow$ solve simultaneous equations
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	$a \sin^3 \theta = 3 \sin^2 \theta \cos \theta$ $\frac{\sin^3 \theta}{\sin^2 \theta \cos \theta} = \frac{3}{a}$ $\frac{\sin \theta}{\cos \theta} = \frac{3}{a}$ $\tan \theta = \frac{3}{a} \quad (\text{shown})$ Since $\tan \theta = \sqrt{2}$, $\therefore \sqrt{2} = \frac{3}{a}$ $a = \frac{3}{\sqrt{2}}$
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